

Quiz 2 — Solutions

M 375T Fall 2026 — Q1: 4 pts; Q2: 6 pts.

1. (i) $z = (8 - 6)/2 = 1$; (ii) mean 0, standard deviation 1 — that is what standardizing does; (iii) $x = \hat{\mu} + \hat{\sigma}z = 6 + 2 \cdot (-1.5) = 3$. *Scoring: 1 pt for (i), 2 pts for (ii) (one each), 1 pt for (iii).* [4 pts]

2. (a) (50000,) — one entry per experiment; each entry counts how many of the 100 uniform samples in that row land in $[0, \frac{1}{2})$. (b) It estimates $\mathbb{P}(\text{exactly 50 of 100 samples land left}) = \binom{100}{50} 2^{-100}$ (≈ 0.0796 ; the expression suffices). (c) ≈ 0.5 : each sample lands left with probability $\frac{1}{2}$, so the average count is ≈ 50 and dividing by 100 gives $\approx \frac{1}{2}$. *Scoring: 2 pts per part — in (a) one for the shape, one for the sentence; in (b) the exact expression is required for both points (the phrase alone earns 1); in (c) one for the value, one for the reason.* [6 pts]